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FACILITATING USER ACTIONS FOR VIRTUAL INTERACTIONS

Abstract

Example methods and systems facilitate user actions during a chat session on an online chat platform. A client device is installed with a chat and video conference application. The chat and video conference application includes a machine learning (ML) model. The client device receives a chat message during a chat session and identifies an action item from the chat message using the ML model. One or more GUI elements can be generated associated with a functionality of an application corresponding to the action item. A GUI element can be activated to invoke the functionality of the application.

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Background/Summary

RELATED APPLICATIONS [0001] The present application is a continuation of U.S. application Ser. No. 18/139,059, filed Apr. 25, 2023, entitled, “Facilitating User Actions for Virtual Interactions,” the entirety of which is incorporated by reference herein.

FIELD

[0002] The present application generally relates to online chat channels and more specifically relates to facilitating user actions for virtual interactions on online chat channels.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] The accompanying drawings, which are incorporated into and constitute a part of this specification, illustrate one or more certain examples and, together with the description of the example, serve to explain the principles and implementations of the certain examples.

[0004] FIG. 1 shows an example system that provides videoconferencing and chat functionality to various client devices;

[0005] FIG. 2 shows an example system in which a chat and video conference provider provides videoconferencing and chat functionality to various client device;

[0006] FIG. 3 shows an example system that facilitate user actions for virtual interactions;

[0007] FIG. 4 shows an example system that is configured to facilitate user actions during a virtual interaction;

[0008] FIG. 5 shows an example master chat panel;

[0009] FIG. 6 shows an example master chat panel displaying GUI elements and suggested reply message associated with an identified action item;

[0010] FIG. 7 shows an example master chat panel displaying GUI elements with an identified action item in the chat window;

[0011] FIG. 8 shows an example master chat panel displaying a chat summary message in the chat window of the design team chat channel;

[0012] FIG. 9 shows an example master chat panel displaying a chat summary message in an external application page;

[0013] FIG. 10 shows an example method for facilitating user actions during a chat session;

[0014] FIG. 11 shows an example computing device suitable for use in example systems or methods for facilitating user actions for virtual interactions, according to certain examples.

DETAILED DESCRIPTION

[0015] Examples are described herein in the context of facilitating user actions for virtual interactions. Those of ordinary skill in the art will realize that the following description is illustrative only and is not intended to be in any way limiting. Reference will now be made in detail to implementations of examples as illustrated in the accompanying drawings. The same reference indicators will be used throughout the drawings and the following description to refer to the same or like items.

[0016] In the interest of clarity, not all of the routine features of the examples described herein are

shown and described. It will, of course, be appreciated that in the development of any such actual implementation, numerous implementation-specific decisions must be made in order to achieve the developer's specific goals, such as compliance with application- and business-related constraints, and that these specific goals will vary from one implementation to another and from one developer to another.

[0017] During virtual interactions, such as online chats, video conferences, or any other online interactions, users may need to switch from one application to another application for certain actions. For example, a user may need to open an online calendar (e.g., Outlook calendar) to check what time the user is available for scheduling an activity with another user while chatting with the other user. However, switching between different applications may be time consuming, interrupt the flow of the virtual interaction, and be prone to error while multitasking. Meanwhile, some actions may not have to be taken during the virtual interaction. However, if the user did not take notes, or could not take notes, the user may forget these to-be-taken actions after the virtual interaction.

[0018] To facilitate user actions during a virtual interaction, the present disclosure provides techniques for generating GUI elements in the GUI of the application for virtual interaction. The GUI elements can be associated with a certain functionality of an external application. Activating a GUI element can invoke the associated functionality of the external application automatically without the user manually launching the external application.

[0019] For example, a first user is chatting with a second user during a chat session on an online chat platform. The first user uses a first client device and the second user uses a second client device. Both the first client device and the second device are installed with a client application provided by the online chat platform. The first client device receives a chat message via the online chat platform. The chat message can be from the first user or the second user. The client application on the first client device uses a machine learning (ML) model, for example a classification model, to determine if the chat message includes an action item based on the chat message and some chat messages before the chat message. If the ML model identifies an action item, the client application then determines which application to execute the action item. The client application then generates a GUI element associated with a functionality of the external application corresponding to the action item. The GUI element is displayed in the GUI of the client application for the chat session with the second user. The first user can activate the GUI element by clicking or touching to invoke the functionality of the external application.

[0020] For example, the chat message is from the second user to the first user that “when are you going to talk with Vijay?” The ML model on the first client device identifies an action item for the first user, that is to check if the first user has a meeting scheduled with Vijay on the first user's calendar. The ML model then accesses the first user's calendar. If the ML model finds that the first user has a meeting scheduled with Vijay at 3 pm on Sunday, the client application on the first client device generates a button with a short description of the meeting. The button is connected to the meeting scheduled on the first user's calendar. The first user can click the button to open the scheduled page on the calendar for verification.

[0021] Meanwhile, the client application can automatically compose a chat message for the first user to the second user. Here, since the ML model finds the first user has a meeting scheduled with Vijay on Sunday, the ML model composes a reply message to the second user's message that “when are you going to talk with Vijay?” The reply message can be composed as “My meeting with Vijay is at 3 pm on Sunday” in the reply panel. The first user can edit it before sending it out.

[0022] In addition, the client application provides a summary of action items after the chat session. The first user may have taken action on some of the action items, but not on other action items. The summary of action items can include a list of action items that the first user has taken action on during the chat session and a list of action items that the user needs to take action on after the chat session. The summary can also include GUI elements for facilitating the first user to take action on

the action items that are not yet taken action on.

[0023] Thus, this example provides GUI elements for facilitating user actions during a chat session. The GUI elements are associated with a functionality of an external application corresponding to an identified action item. The GUI elements are displayed in the GUI of the chat session so that the user can take action quickly and directly without minimizing the client application, launching the external application, invoking a corresponding functionality, etc. to take action on the action item. Meanwhile, a summary of action items can further assist the user and save time. The user does not need to write down any after-session steps during the virtual interaction session. The summary includes all action items so that the user has a full picture of the actions spawn out of the virtual interaction session.

[0024] This illustrative example is given to introduce the reader to the general subject matter discussed herein and the disclosure is not limited to this example. The following sections describe various additional non-limiting examples of facilitating user actions for virtual interactions.

[0025] Referring now to FIG. 1, FIG. 1 shows an example system **100** that provides videoconferencing functionality to various client devices. The system **100** includes a chat and video conference provider **110** that is connected to multiple communication networks **120**, **130**, through which various client devices **140-180** can participate in video conferences hosted by the chat and video conference provider **110**. For example, the chat and video conference provider **110** can be located within a private network to provide video conferencing services to devices within the private network, or it can be connected to a public network, e.g., the internet, so it may be accessed by anyone. Some examples may even provide a hybrid model in which a chat and video conference provider **110** may supply components to enable a private organization to host private internal video conferences or to connect its system to the chat and video conference provider **110** over a public network.

[0026] The system optionally also includes one or more authentication and authorization providers, e.g., authentication and authorization provider **115**, which can provide authentication and authorization services to users of the client devices **140-160**. Authentication and authorization provider **115** may authenticate users to the chat and video conference provider **110** and manage user authorization for the various services provided by chat and video conference provider **110**. In this example, the authentication and authorization provider **115** is operated by a different entity than the chat and video conference provider **110**, though in some examples, they may be the same entity.

[0027] Chat and video conference provider **110** allows clients to create videoconference meetings (or “meetings”) and invite others to participate in those meetings as well as perform other related functionality, such as recording the meetings, generating transcripts from meeting audio, generating summaries and translations from meeting audio, manage user functionality in the meetings, enable text messaging during the meetings, create and manage breakout rooms from the virtual meeting, etc. FIG. 2, described below, provides a more detailed description of the architecture and functionality of the chat and video conference provider **110**. It should be understood that the term “meeting” encompasses the term “webinar” used herein.

[0028] Meetings in this example chat and video conference provider **110** are provided in virtual rooms to which participants are connected. The room in this context is a construct provided by a server that provides a common point at which the various video and audio data is received before being multiplexed and provided to the various participants. While a “room” is the label for this concept in this disclosure, any suitable functionality that enables multiple participants to participate in a common videoconference may be used.

[0029] To create a meeting with the chat and video conference provider **110**, a user may contact the chat and video conference provider **110** using a client device **140-180** and select an option to create a new meeting. Such an option may be provided in a webpage accessed by a client device **140-160** or a client application executed by a client device **140-160**. For telephony devices, the user may be

presented with an audio menu that they may navigate by pressing numeric buttons on their telephony device. To create the meeting, the chat and video conference provider **110** may prompt the user for certain information, such as a date, time, and duration for the meeting, a number of participants, a type of encryption to use, whether the meeting is confidential or open to the public, etc. After receiving the various meeting settings, the chat and video conference provider may create a record for the meeting and generate a meeting identifier and, in some examples, a corresponding meeting password or passcode (or other authentication information), all of which meeting information is provided to the meeting host.

[0030] After receiving the meeting information, the user may distribute the meeting information to one or more users to invite them to the meeting. To begin the meeting at the scheduled time (or immediately, if the meeting was set for an immediate start), the host provides the meeting identifier and, if applicable, corresponding authentication information (e.g., a password or passcode). The video conference system then initiates the meeting and may admit users to the meeting. Depending on the options set for the meeting, the users may be admitted immediately upon providing the appropriate meeting identifier (and authentication information, as appropriate), even if the host has not yet arrived, or the users may be presented with information indicating that the meeting has not yet started, or the host may be required to specifically admit one or more of the users.

[0031] During the meeting, the participants may employ their client devices **140-180** to capture audio or video information and stream that information to the chat and video conference provider **110**. They also receive audio or video information from the chat and video conference provider **110**, which is displayed by the respective client device **140** to enable the various users to participate in the meeting.

[0032] At the end of the meeting, the host may select an option to terminate the meeting, or it may terminate automatically at a scheduled end time or after a predetermined duration. When the meeting terminates, the various participants are disconnected from the meeting, and they will no longer receive audio or video streams for the meeting (and will stop transmitting audio or video streams). The chat and video conference provider **110** may also invalidate the meeting information, such as the meeting identifier or password/passcode.

[0033] To provide such functionality, one or more client devices **140-180** may communicate with the chat and video conference provider **110** using one or more communication networks, such as network **120** or the public switched telephone network (“PSTN”) **130**. The client devices **140-180** may be any suitable computing or communication devices that have audio or video capability. For example, client devices **140-160** may be conventional computing devices, such as desktop or laptop computers having processors and computer-readable media, connected to the chat and video conference provider **110** using the internet or other suitable computer network. Suitable networks include the internet, any local area network (“LAN”), metro area network (“MAN”), wide area network (“WAN”), cellular network (e.g., 3G, 4G, 4G LTE, 5G, etc.), or any combination of these. Other types of computing devices may be used instead or as well, such as tablets, smartphones, and dedicated video conferencing equipment. Each of these devices may provide both audio and video capabilities and may enable one or more users to participate in a video conference meeting hosted by the chat and video conference provider **110**.

[0034] In addition to the computing devices discussed above, client devices **140-180** may also include one or more telephony devices, such as cellular telephones (e.g., cellular telephone **170**), internet protocol (“IP”) phones (e.g., telephone **180**), or conventional telephones. Such telephony devices may allow a user to make conventional telephone calls to other telephony devices using the PSTN, including the chat and video conference provider **110**. It should be appreciated that certain computing devices may also provide telephony functionality and may operate as telephony devices. For example, smartphones typically provide cellular telephone capabilities and thus may operate as telephony devices in the example system **100** shown in FIG. **1**. In addition, conventional computing devices may execute software to enable telephony functionality, which may allow the user to make

and receive phone calls, e.g., using a headset and microphone. Such software may communicate with a PSTN gateway to route the call from a computer network to the PSTN. Thus, telephony devices encompass any devices that can make conventional telephone calls and are not limited solely to dedicated telephony devices like conventional telephones.

[0035] Referring again to client devices **140-160**, these devices **140-160** contact the chat and video conference provider **110** using network **120** and may provide information to the chat and video conference provider **110** to access functionality provided by the chat and video conference provider **110**, such as access to create new meetings or join existing meetings. To do so, the client devices **140-160** may provide user authentication information, meeting identifiers, meeting passwords or passcodes, etc. In examples that employ an authentication and authorization provider **115**, a client device, e.g., client devices **140-160**, may operate in conjunction with an authentication and authorization provider **115** to provide authentication and authorization information or other user information to the chat and video conference provider **110**.

[0036] An authentication and authorization provider **115** may be any entity trusted by the chat and video conference provider **110** that can help authenticate a user to the chat and video conference provider **110** and authorize the user to access the services provided by the chat and video conference provider **110**. For example, a trusted entity may be a server operated by a business or other organization with whom the user has created an account, including authentication and authorization information, such as an employer or trusted third-party. The user may sign into the authentication and authorization provider **115**, such as by providing a username and password, to access their account information at the authentication and authorization provider **115**. The account information includes information established and maintained at the authentication and authorization provider **115** that can be used to authenticate and facilitate authorization for a particular user, irrespective of the client device they may be using. An example of account information may be an email account established at the authentication and authorization provider **115** by the user and secured by a password or additional security features, such as single sign-on, hardware tokens, two-factor authentication, etc. However, such account information may be distinct from functionality such as email. For example, a health care provider may establish accounts for its patients. And while the related account information may have associated email accounts, the account information is distinct from those email accounts.

[0037] Thus, a user's account information relates to a secure, verified set of information that can be used to authenticate and provide authorization services for a particular user and should be accessible only by that user. By properly authenticating, the associated user may then verify themselves to other computing devices or services, such as the chat and video conference provider **110**. The authentication and authorization provider **115** may require the explicit consent of the user before allowing the chat and video conference provider **110** to access the user's account information for authentication and authorization purposes.

[0038] Once the user is authenticated, the authentication and authorization provider **115** may provide the chat and video conference provider **110** with information about services the user is authorized to access. For instance, the authentication and authorization provider **115** may store information about user roles associated with the user. The user roles may include collections of services provided by the chat and video conference provider **110** that users assigned to those user roles are authorized to use. Alternatively, more or less granular approaches to user authorization may be used.

[0039] When the user accesses the chat and video conference provider **110** using a client device, the chat and video conference provider **110** communicates with the authentication and authorization provider **115** using information provided by the user to verify the user's account information. For example, the user may provide a username or cryptographic signature associated with an authentication and authorization provider **115**. The authentication and authorization provider **115** then either confirms the information presented by the user or denies the request. Based on this

response, the chat and video conference provider **110** either provides or denies access to its services, respectively.

[0040] For telephony devices, e.g., client devices **170-180**, the user may place a telephone call to the chat and video conference provider **110** to access video conference services. After the call is answered, the user may provide information regarding a video conference meeting, e.g., a meeting identifier (“ID”), a passcode or password, etc., to allow the telephony device to join the meeting and participate using audio devices of the telephony device, e.g., microphone(s) and speaker(s), even if video capabilities are not provided by the telephony device.

[0041] Because telephony devices typically have more limited functionality than conventional computing devices, they may be unable to provide certain information to the chat and video conference provider **110**. For example, telephony devices may be unable to provide authentication information to authenticate the telephony device or the user to the chat and video conference provider **110**. Thus, the chat and video conference provider **110** may provide more limited functionality to such telephony devices. For example, the user may be permitted to join a meeting after providing meeting information, e.g., a meeting identifier and passcode, but only as an anonymous participant in the meeting. This may restrict their ability to interact with the meetings in some examples, such as by limiting their ability to speak in the meeting, hear or view certain content shared during the meeting, or access other meeting functionality, such as joining breakout rooms or engaging in text chat with other participants in the meeting.

[0042] It should be appreciated that users may choose to participate in meetings anonymously and decline to provide account information to the chat and video conference provider **110**, even in cases where the user could authenticate and employs a client device capable of authenticating the user to the chat and video conference provider **110**. The chat and video conference provider **110** may determine whether to allow such anonymous users to use services provided by the chat and video conference provider **110**. Anonymous users, regardless of the reason for anonymity, may be restricted as discussed above with respect to users employing telephony devices, and in some cases may be prevented from accessing certain meetings or other services, or may be entirely prevented from accessing the chat and video conference provider **110**.

[0043] Referring again to chat and video conference provider **110**, in some examples, it may allow client devices **140-160** to encrypt their respective video and audio streams to help improve privacy in their meetings. Encryption may be provided between the client devices **140-160** and the chat and video conference provider **110** or it may be provided in an end-to-end configuration where multimedia streams (e.g., audio or video streams) transmitted by the client devices **140-160** are not decrypted until they are received by another client device **140-160** participating in the meeting. Encryption may also be provided during only a portion of a communication, for example encryption may be used for otherwise unencrypted communications that cross international borders.

[0044] Client-to-server encryption may be used to secure the communications between the client devices **140-160** and the chat and video conference provider **110**, while allowing the chat and video conference provider **110** to access the decrypted multimedia streams to perform certain processing, such as recording the meeting for the participants or generating transcripts of the meeting for the participants. End-to-end encryption may be used to keep the meeting entirely private to the participants without any worry about a chat and video conference provider **110** having access to the substance of the meeting. Any suitable encryption methodology may be employed, including key-pair encryption of the streams. For example, to provide end-to-end encryption, the meeting host's client device may obtain public keys for each of the other client devices participating in the meeting and securely exchange a set of keys to encrypt and decrypt multimedia content transmitted during the meeting. Thus, the client devices **140-160** may securely communicate with each other during the meeting. Further, in some examples, certain types of encryption may be limited by the types of devices participating in the meeting. For example, telephony devices may lack the ability

to encrypt and decrypt multimedia streams. Thus, while encrypting the multimedia streams may be desirable in many instances, it is not required as it may prevent some users from participating in a meeting.

[0045] By using the example system shown in FIG. 1, users can create and participate in meetings using their respective client devices **140-180** via the chat and video conference provider **110**.

Further, such a system enables users to use a wide variety of different client devices **140-180** from traditional standards-based video conferencing hardware to dedicated video conferencing equipment to laptop or desktop computers to handheld devices to legacy telephony devices, etc.

[0046] Referring now to FIG. 2, FIG. 2 shows an example system **200** in which a chat and video conference provider **210** provides videoconferencing functionality to various client devices **220-250**. The client devices **220-250** include two conventional computing devices **220-230**, dedicated equipment for a video conference room **240**, and a telephony device **250**. Each client device **220-250** communicates with the chat and video conference provider **210** over a communications network, such as the internet for client devices **220-240** or the PSTN for client device **250**, generally as described above with respect to FIG. 1. The chat and video conference provider **210** is also in communication with one or more authentication and authorization providers **215**, which can authenticate various users to the chat and video conference provider **210** generally as described above with respect to FIG. 1.

[0047] In this example, the chat and video conference provider **210** employs multiple different servers (or groups of servers) to provide different examples of video conference functionality, thereby enabling the various client devices to create and participate in video conference meetings. The chat and video conference provider **210** uses one or more real-time media servers **212**, one or more network services servers **214**, one or more video room gateways **216**, one or more message and presence gateways **217**, and one or more telephony gateways **218**. Each of these servers **212-218** is connected to one or more communications networks to enable them to collectively provide access to and participation in one or more video conference meetings to the client devices **220-250**.

[0048] The real-time media servers **212** provide multiplexed multimedia streams to meeting participants, such as the client devices **220-250** shown in FIG. 2. While video and audio streams typically originate at the respective client devices, they are transmitted from the client devices **220-250** to the chat and video conference provider **210** via one or more networks where they are received by the real-time media servers **212**. The real-time media servers **212** determine which protocol is optimal based on, for example, proxy settings and the presence of firewalls, etc. For example, the client device might select among UDP, TCP, TLS, or HTTPS for audio and video and UDP for content screen sharing.

[0049] The real-time media servers **212** then multiplex the various video and audio streams based on the target client device and communicate multiplexed streams to each client device. For example, the real-time media servers **212** receive audio and video streams from client devices **220-240** and only an audio stream from client device **250**. The real-time media servers **212** then multiplex the streams received from devices **230-250** and provide the multiplexed stream to client device **220**. The real-time media servers **212** are adaptive, for example, reacting to real-time network and client changes, in how they provide these streams. For example, the real-time media servers **212** may monitor parameters such as a client's bandwidth CPU usage, memory and network I/() as well as network parameters such as packet loss, latency and jitter to determine how to modify the way in which streams are provided.

[0050] The client device **220** receives the stream, performs any decryption, decoding, and demultiplexing on the received streams, and then outputs the audio and video using the client device's video and audio devices. In this example, the real-time media servers do not multiplex client device **220**'s own video and audio feeds when transmitting streams to it. Instead, each client device **220-250** only receives multimedia streams from other client devices **220-250**. For telephony devices that lack video capabilities, e.g., client device **250**, the real-time media servers **212** only

deliver multiplex audio streams. The client device **220** may receive multiple streams for a particular communication, allowing the client device **220** to switch between streams to provide a higher quality of service.

[0051] In addition to multiplexing multimedia streams, the real-time media servers **212** may also decrypt incoming multimedia stream in some examples. As discussed above, multimedia streams may be encrypted between the client devices **220-250** and the chat and video conference provider **210**. In some such examples, the real-time media servers **212** may decrypt incoming multimedia streams, multiplex the multimedia streams appropriately for the various clients, and encrypt the multiplexed streams for transmission.

[0052] As mentioned above with respect to FIG. **1**, the chat and video conference provider **210** may provide certain functionality with respect to unencrypted multimedia streams at a user's request. For example, the meeting host may be able to request that the meeting be recorded or that a transcript of the audio streams be prepared, which may then be performed by the real-time media servers **212** using the decrypted multimedia streams, or the recording or transcription functionality may be off-loaded to a dedicated server (or servers), e.g., cloud recording servers, for recording the audio and video streams. In some examples, the chat and video conference provider **210** may allow a meeting participant to notify it of inappropriate behavior or content in a meeting. Such a notification may trigger the real-time media servers to **212** record a portion of the meeting for review by the chat and video conference provider **210**. Still other functionality may be implemented to take actions based on the decrypted multimedia streams at the chat and video conference provider, such as monitoring video or audio quality, adjusting or changing media encoding mechanisms, etc.

[0053] It should be appreciated that multiple real-time media servers **212** may be involved in communicating data for a single meeting and multimedia streams may be routed through multiple different real-time media servers **212**. In addition, the various real-time media servers **212** may not be co-located, but instead may be located at multiple different geographic locations, which may enable high-quality communications between clients that are dispersed over wide geographic areas, such as being located in different countries or on different continents. Further, in some examples, one or more of these servers may be co-located on a client's premises, e.g., at a business or other organization. For example, different geographic regions may each have one or more real-time media servers **212** to enable client devices in the same geographic region to have a high-quality connection into the chat and video conference provider **210** via local servers **212** to send and receive multimedia streams, rather than connecting to a real-time media server located in a different country or on a different continent. The local real-time media servers **212** may then communicate with physically distant servers using high-speed network infrastructure, e.g., internet backbone network(s), that otherwise might not be directly available to client devices **220-250** themselves. Thus, routing multimedia streams may be distributed throughout the video conference system and across many different real-time media servers **212**.

[0054] Turning to the network services servers **214**, these servers **214** provide administrative functionality to enable client devices to create or participate in meetings, send meeting invitations, create or manage user accounts or subscriptions, and other related functionality. Further, these servers may be configured to perform different functionalities or to operate at different levels of a hierarchy, e.g., for specific regions or localities, to manage portions of the chat and video conference provider under a supervisory set of servers. When a client device **220-250** accesses the chat and video conference provider **210**, it will typically communicate with one or more network services servers **214** to access their account or to participate in a meeting.

[0055] When a client device **220-250** first contacts the chat and video conference provider **210** in this example, it is routed to a network services server **214**. The client device may then provide access credentials for a user, e.g., a username and password or single sign-on credentials, to gain authenticated access to the chat and video conference provider **210**. This process may involve the

network services servers **214** contacting an authentication and authorization provider **215** to verify the provided credentials. Once the user's credentials have been accepted, and the user has consented, the network services servers **214** may perform administrative functionality, like updating user account information, if the user has account information stored with the chat and video conference provider **210**, or scheduling a new meeting, by interacting with the network services servers **214**. Authentication and authorization provider **215** may be used to determine which administrative functionality a given user may access according to assigned roles, permissions, groups, etc.

[0056] In some examples, users may access the chat and video conference provider **210** anonymously. When communicating anonymously, a client device **220-250** may communicate with one or more network services servers **214** but only provide information to create or join a meeting, depending on what features the chat and video conference provider allows for anonymous users. For example, an anonymous user may access the chat and video conference provider using client device **220** and provide a meeting ID and passcode. The network services server **214** may use the meeting ID to identify an upcoming or on-going meeting and verify the passcode is correct for the meeting ID. After doing so, the network services server(s) **214** may then communicate information to the client device **220** to enable the client device **220** to join the meeting and communicate with appropriate real-time media servers **212**.

[0057] In cases where a user wishes to schedule a meeting, the user (anonymous or authenticated) may select an option to schedule a new meeting and may then select various meeting options, such as the date and time for the meeting, the duration for the meeting, a type of encryption to be used, one or more users to invite, privacy controls (e.g., not allowing anonymous users, preventing screen sharing, manually authorize admission to the meeting, etc.), meeting recording options, etc. The network services servers **214** may then create and store a meeting record for the scheduled meeting. When the scheduled meeting time arrives (or within a threshold period of time in advance), the network services server(s) **214** may accept requests to join the meeting from various users.

[0058] To handle requests to join a meeting, the network services server(s) **214** may receive meeting information, such as a meeting ID and passcode, from one or more client devices **220-250**. The network services server(s) **214** locate a meeting record corresponding to the provided meeting ID and then confirm whether the scheduled start time for the meeting has arrived, whether the meeting host has started the meeting, and whether the passcode matches the passcode in the meeting record. If the request is made by the host, the network services server(s) **214** activates the meeting and connects the host to a real-time media server **212** to enable the host to begin sending and receiving multimedia streams.

[0059] Once the host has started the meeting, subsequent users requesting access will be admitted to the meeting if the meeting record is located and the passcode matches the passcode supplied by the requesting client device **220-250**. In some examples additional access controls may be used as well. But if the network services server(s) **214** determines to admit the requesting client device **220-250** to the meeting, the network services server **214** identifies a real-time media server **212** to handle multimedia streams to and from the requesting client device **220-250** and provides information to the client device **220-250** to connect to the identified real-time media server **212**. Additional client devices **220-250** may be added to the meeting as they request access through the network services server(s) **214**.

[0060] After joining a meeting, client devices will send and receive multimedia streams via the real-time media servers **212**, but they may also communicate with the network services servers **214** as needed during meetings. For example, if the meeting host leaves the meeting, the network services server(s) **214** may appoint another user as the new meeting host and assign host administrative privileges to that user. Hosts may have administrative privileges to allow them to manage their meetings, such as by enabling or disabling screen sharing, muting or removing users

from the meeting, assigning or moving users to the mainstage or a breakout room if present, recording meetings, etc. Such functionality may be managed by the network services server(s) **214**. [0061] For example, if a host wishes to remove a user from a meeting, they may select a user to remove and issue a command through a user interface on their client device. The command may be sent to a network services server **214**, which may then disconnect the selected user from the corresponding real-time media server **212**. If the host wishes to remove one or more participants from a meeting, such a command may also be handled by a network services server **214**, which may terminate the authorization of the one or more participants for joining the meeting.

[0062] In addition to creating and administering on-going meetings, the network services server(s) **214** may also be responsible for closing and tearing-down meetings once they have been completed. For example, the meeting host may issue a command to end an on-going meeting, which is sent to a network services server **214**. The network services server **214** may then remove any remaining participants from the meeting, communicate with one or more real time media servers **212** to stop streaming audio and video for the meeting, and deactivate, e.g., by deleting a corresponding passcode for the meeting from the meeting record, or delete the meeting record(s) corresponding to the meeting. Thus, if a user later attempts to access the meeting, the network services server(s) **214** may deny the request.

[0063] Depending on the functionality provided by the chat and video conference provider, the network services server(s) **214** may provide additional functionality, such as by providing private meeting capabilities for organizations, special types of meetings (e.g., webinars), etc. Such functionality may be provided according to various examples of video conferencing providers according to this description.

[0064] Referring now to the video room gateway servers **216**, these servers **216** provide an interface between dedicated video conferencing hardware, such as may be used in dedicated video conferencing rooms. Such video conferencing hardware may include one or more cameras and microphones and a computing device designed to receive video and audio streams from each of the cameras and microphones and connect with the chat and video conference provider **210**. For example, the video conferencing hardware may be provided by the chat and video conference provider to one or more of its subscribers, which may provide access credentials to the video conferencing hardware to use to connect to the chat and video conference provider **210**.

[0065] The video room gateway servers **216** provide specialized authentication and communication with the dedicated video conferencing hardware that may not be available to other client devices **220-230, 250**. For example, the video conferencing hardware may register with the chat and video conference provider when it is first installed and the video room gateway may authenticate the video conferencing hardware using such registration as well as information provided to the video room gateway server(s) **216** when dedicated video conferencing hardware connects to it, such as device ID information, subscriber information, hardware capabilities, hardware version information etc. Upon receiving such information and authenticating the dedicated video conferencing hardware, the video room gateway server(s) **216** may interact with the network services servers **214** and real-time media servers **212** to allow the video conferencing hardware to create or join meetings hosted by the chat and video conference provider **210**.

[0066] Referring now to the telephony gateway servers **218**, these servers **218** enable and facilitate telephony devices' participation in meetings hosted by the chat and video conference provider **210**. Because telephony devices communicate using the PSTN and not using computer networking protocols, such as TCP/IP, the telephony gateway servers **218** act as an interface that converts between the PSTN, and the networking system used by the chat and video conference provider **210**.

[0067] For example, if a user uses a telephony device to connect to a meeting, they may dial a phone number corresponding to one of the chat and video conference provider's telephony gateway servers **218**. The telephony gateway server **218** will answer the call and generate audio messages requesting information from the user, such as a meeting ID and passcode. The user may enter such

information using buttons on the telephony device, e.g., by sending dual-tone multi-frequency (“DTMF”) audio streams to the telephony gateway server **218**. The telephony gateway server **218** determines the numbers or letters entered by the user and provides the meeting ID and passcode information to the network services servers **214**, along with a request to join or start the meeting, generally as described above. Once the telephony client device **250** has been accepted into a meeting, the telephony gateway server is instead joined to the meeting on the telephony device's behalf.

[0068] After joining the meeting, the telephony gateway server **218** receives an audio stream from the telephony device and provides it to the corresponding real-time media server **212** and receives audio streams from the real-time media server **212**, decodes them, and provides the decoded audio to the telephony device. Thus, the telephony gateway servers **218** operate essentially as client devices, while the telephony device operates largely as an input/output device, e.g., a microphone and speaker, for the corresponding telephony gateway server **218**, thereby enabling the user of the telephony device to participate in the meeting despite not using a computing device or video.

[0069] It should be appreciated that the components of the chat and video conference provider **210** discussed above are merely examples of such devices and an example architecture. Some video conference providers may provide more or less functionality than described above and may not separate functionality into different types of servers as discussed above. Instead, any suitable servers and network architectures may be used according to different examples.

[0070] Referring now to FIG. 3, FIG. 3 shows an example system **300** that facilitate user actions for virtual interactions. In this example system **300**, a chat and video conference provider **310** and a number of client device **340A-340N** (which may be referred to herein individually as a client device **340** or collectively as the client devices **340**) are connected via a network **320**. The chat and video conference provider **310** can be the chat and video conference provider **110** in FIG. 1 or the chat and video conference provider **210** in FIG. 2. The network **320** can be the internet or any suitable communications network or combination of communications network may be employed, including LANs (e.g., within a corporate private LAN), WANs, MANs, cellular network (e.g., 3G, 4G, 4G LTE, 5G, etc.), or any combination of these.

[0071] The client devices **340** can be any suitable computing or communications device. The client device **340** can be a client device (e.g., **140**, **150**, **160**, or **170**) in FIG. 1 or a client device (e.g., **220**, **230**, or **250**) in FIG. 2. For example, client devices **340** may be desktop computers, laptop computers, tablets, smart phones having processors and computer-readable media, connected to the chat and video conference provider **310** using the internet or other suitable computer network. The client devices **340** have chat and video conference software installed to enable them to connect to the chat and video conference provider **310**. During a chat session, a user associated a client device (e.g., client device **340A**) can interact with other users associated with other client devices (e.g., client device **340B-340N**) via the chat and video conference provider **310** by sending and receiving chat messages, and reacting to received chat messages.

[0072] Now referring to FIG. 4, FIG. 4 shows an example system **400** that is configured to facilitate user actions during a virtual interaction. The chat and video conference provider **310** is in network communication with a client device **340**. The client device **340** is installed with a chat and video conference application provided by the chat and video conference provider **310**.

[0073] The chat and video conference provider **310** includes a data store **410** and a model store **420**. The data store **410** stores historical chat data from different client devices **340**. The historical chat data can include message identifications, channel identifications, parent message identification (if any), sender identification, message content, send time, reply to the message (if any), and reply time (if any). The data store **410** can also store user metadata. The user metadata includes user profile data, for example, title, team, location, chat channels, chat contacts, and any information included in the user profile on the chat and video conference platform. The user metadata also includes user activity data, such as what actions the user takes after each chat message (e.g.,

checking online calendar, scheduling video meeting, etc.).

[0074] The model store **420** includes different AI/ML models for facilitating user actions during a virtual interaction. Various types of models or artificial intelligence algorithms may be used in example systems. For example, simple machine learning models, such as Linear Regression and Gradient Boosting may be used. In other examples, more sophisticated models, such as Factorization Machines (“FM”). As more data is available in a system according to these examples, deep learning models may be utilized, such as DeepFM and Wide&Deep or other similar models. Other alternative machine-learning models that might be used include a deep convolutional neural network (“CNN”), a residual neural network (“Resnet”), or a recurrent neural network, e.g., long short-term memory (“LSTM”) models or gated recurrent units (“GRUs”) models. The machine-learning model can also be any other suitable machine-learning model, such as a three-dimensional CNN (“3DCNN”), a dynamic time warping (“DTW”) technique, a hidden Markov model (“HMM”), etc., or combinations of one or more of such techniques—e.g., CNN-HMM or MCNN (Multi-Scale Convolutional Neural Network). Further, some examples may employ adversarial networks, such as generative adversarial networks (“GANs”), or may employ autoencoders (“AEs”) in conjunction with machine-learning models, such as AEGANs or variational AEGANs (“VAEGANs”). Alternatively, some machine-learning models may use transformer networks or self-attention based neural networks. In addition, the models or artificial intelligence algorithms in the model store can be supervised or unsupervised learning models.

[0075] In some examples, the chat and video conference provider **310** is configured to select an ML model from the model store **330** and train the ML model for classification. Classification can be a task for matching an item with a class or group based on natural language processing. The classification model can be based on logistic regression, decision tree, random forest, support vector machine, K-nearest neighbor, naive Bayes, stochastic gradient descent, or any suitable ML algorithms for classification.

[0076] During the training phase, the chat and video conference provider **310** receives a training dataset. In some examples, the training dataset include historical chat messages and action item labels. Certain features can be derived from the historical chat messages for classifying the corresponding chat messages into certain action categories. some of the features may include keywords associated with certain user actions. For example, time related words may be associated with an action on the calendar. Also, for example, “schedule a meeting” or alternative phrases may trigger an action for creating a virtual meeting. In some examples, the training dataset includes historical chat messages and user metadata. The historical chat messages are not labeled with any action item labels. The user metadata can include user profile data, for example, title, team, location, chat channels, chat contacts, and any information included in the user profile on the chat and video conference platform. The user metadata can also include user activity data, for example, user actions associated with the chat messages and user interactions with other applications on the chat and video conference platform. The chat and video conference provider **310** can train the ML model to extract features from the historical chat messages and identify action items to create action categories. The action categories can be encoded in the classification model. The action categories correspond to a functionality of an application that can execute the corresponding action items.

[0077] In some examples, the chat and video conference provider **310** trains and stores multiple classification models for different user groups or chat channels. The chat and video conference provider **310** can generate different user groups based on the user profile data stored in the data store **410**. Certain user groups may have their frequent action items appear in their chat messages. Also, certain chat channels, especially chat channels based on work groups, can have their preferred action items.

[0078] In some examples, the chat and video conference provider **310** includes an action facilitation engine **430** configured to determine action items in chat messages and determine an

internal or external application to execute the action items. The internal application can be another application provided by the chat and video conference provider **310**, other than the chat and video conference application **450**. The external application can be an application provided by a third-party provider. Either the internal application or the external application can be integrated with the chat and video conference application **450**. The action facilitation engine **430** may implement an ML model that is not tailored to any specific user but apply to various users. In some examples, the action facilitation engine **430** implements a trained ML model for action facilitation stored in the model store **420** corresponding to the user group that the user belongs to. Alternative, the action facilitation engine **430** can include the multiple trained ML models for action facilitation from the model store **420** and select one trained model based on the user group of the user to facilitate the user. In some examples, the ML model includes a map between action categories and functionalities of applications or applications that can execute action items in corresponding action categories. Even though in this example an ML model is used in for action facilitation, the action facilitation engine **430** is not limited to using an ML model for identifying action items and determining external applications to execute the action items. The action facilitation engine **430** can implement any suitable models or algorithms.

[0079] The action facilitation engine **430** on the chat and video conference provider **310** is not limited to identify action items in chat messages and determine applications to execute the identified action items for chat session. The action facilitation engine **430** can also identify action items during video conferences, which can be triggered from in-meeting real-time speech transcriptions. Names of the participants in a virtual conference can be available to the video conference provider **310**. The action facilitation engine **430** can automatically tag the relevant participant with suggested actions.

[0080] The chat and video conference application **450** installed on the client device **340** includes a user facilitation engine **460** and a local data store **470**. The local data store **470** stores chat data and user metadata for the user associated with the client device **440**. The local data store **380** also stores feedback input from the user regarding the accuracy of the identified action items. In some examples, the user facilitation engine **460** is part of the chat and video conference application **450**. In some examples, the user facilitation engine **460** is an independent application integrated into the chat and video conference application **350** as microservices. The independent application can be provided by the chat and video conference provider **310**. Alternatively, the independent application can be provided by a third-party provider.

[0081] In some examples, the chat and video conference application **450** includes the user facilitation engine **460** for identifying action items from chat messages and determining applications to execute corresponding action items, even if the chat and video conference provider **310** includes an action facilitation engine **430** configured with the same functionality. In some examples, the chat and video conference provider **310** selects a trained ML model for action facilitation to be initially implemented by the user facilitation engine **460**. For example, the chat and video conference provider **310** can determine a user group for the user associated with the client device **340** based on user profile data. The chat and video conference provider **310** then transmits a trained ML model corresponding to the user group to the chat and video conference application **450** installed on the client device **340**. The chat and video conference application **450** can implement the trained ML model for action facilitation corresponding to the user group in the user facilitation engine **460**.

[0082] When the chat and video conference application **450** is just installed on the client device **340**, few data for the user associated with the client device **340** is collected or stored at the local data store **480**. As more chat data, user metadata, and feedback input from the user associated with client device **340** is collected, the ML model implemented in the user facilitation engine **460** can learn and tune user specific features and corresponding action categories to more accurately identify action items to the user. Even though in this example an ML-based model is used for

building the classification model implemented in the user facilitation engine **460**, the user facilitation engine **460** is not limited to using an ML model for predicting action items from chat messages. The user facilitation engine **460** can implement any suitable models or algorithms. [0083] The chat and video conference application **450** is integrated with other applications associated with action items identified by the user facilitation engine **460**. The user facilitation engine **460** is also configured to generate a GUI element associated with a functionality of a second application corresponding to an identified action item. The second application can be another application integrated with the chat and video conference application **450**. Alternatively, or additionally, the second application is the same chat and video conference application **450** which includes a different functionality besides chat, for example video conferencing, phone calls, etc. In some examples, the user facilitation engine **460** does not include an ML model or other algorithm for identifying action items from chat messages and determining applications to execute corresponding action items. The action facilitation engine **430** on the chat and video conference provider **310** can identify the action items and determine the applications to execute corresponding action items, and transmit such information to the user facilitation engine **460** on the client device **340**. The user facilitation engine **460** can generate GUI elements associated with the applications to execute corresponding action items. The chat and video conference application **450** can provide a GUI for participants to interact with each other via the chat and video conference provider **310**. The GUI elements associated with other applications to execute action items can be displayed in the GUI of the chat and video conference application **450**. In some examples, virtual conferences between participants are established via the GUI of the chat and video conference application **450**. The GUI may include components for scraping video signals during video conferences, for example performing Optical Character Recognition (OCR) to detect and extract text within video streams, including participant names and materials shown on the screen (e.g., presentation slides, histogram chart, etc.). Video scraping can enhance action facilitation by disambiguating entities in action.

[0084] Now referring to FIG. 5, FIG. 5 shows an example master chat panel **500**. The master chat panel **500** may be displayed on a client device **340** in response to information sent by a chat and video conference provider, such as the chat and video conference provider **310** in FIG. 3. The master chat panel **500** may be generated by a chat application run by one or more processors stored on the client device **340**. In some examples, the application is a standalone chat client. In some examples, the application is integrated into a video conferencing application, such as a chat and video conference application **450** in FIG. 4. The client device **340** may be a personal computer such as a desktop or laptop, a mobile device, or other computing device having at least a processor, non-transitory memory, a user input system, and communication interface, providing network connectivity.

[0085] The master chat panel **500** may include a general dashboard **505**, a chat control dashboard **545**, a sidebar **515**, a chat window **550**, a reply dashboard **560**, and a reply panel **555**. The general dashboard **505** may include one or more buttons or links that switch functionalities and/or views of the master chat panel **500**. For example, FIG. 5 shows a chat view, perhaps in response to a user command selecting a chat button **510** in the general dashboard **505**. In this view, the chat window **550**, the reply panel **555**, and other components illustrated in FIG. 5 may be displayed on the client device **340**. In other examples, a contacts button may be selected by a user. In response the contacts button being selected, the chat window **550**, the reply dashboard **560** and the reply panel **555** may be replaced by a display of a contacts window including a list of user contacts associated with the user of the client device. The sidebar **515** may be displayed alongside the contacts window. Other configurations are also possible. Various buttons on the general dashboard **505** may correspond to various displays of windows being displayed on the client device **340**. Any number of components shown in FIG. 5 may be displayed on the client device **340** with any of the various windows. Similarly, any of the components may cease to be displayed in accordance with any of the

windows.

[0086] The sidebar **515** may include one or more chat channel headings. A chat heading may include one or more chat channels such as the chat channel **504a-b**. In some examples, the chat channel may include a private chat **504a**, where the chat is between the user associated with the client device **340** and another user. Messages sent and received via the chat channel **504a** may only be accessed by the users in the chat channel **504a**. Thus, the client devices **340** associated with the user and the client device **340** associated with the other user may securely communicate with each other. In some examples, the chat channel may be a group chat **504b**, where two or more users have access to send and receive messages within the chat channel. In some examples, the chat channel **504b** may only be accessed by users who have permission to enter the chat channel. A host of the chat channel **504b** and/or the chat and video conference provider **310** may grant access to the chat channel **504b**. Although only the chat channel headings including chat channels **504a-b** are shown, other chat channel headings are possible. For example, some examples may include a chat channel heading that displays, on the client device **340**, only those channels that the user associated with the client device **340** is a member of that have been recently accessed. “Recently accessed” may be determined by the client device **340** to be a fixed number of most recent channels accessed by the user, or may be only those channels accessed within a certain time, calculated from the current time.

[0087] The sidebar **515** may include an application heading. The application heading may include one or more independent applications integrated into the chat application (e.g., chat and video conference application **450**) as microservices. The independent applications can be provided by the chat and video conference provider **310**. Alternatively, or additionally, the independent applications can be provided by a third-party provider. In some examples, the one or more independent applications include a facilitation application **508** configured to identify action items from chat messages, for example using the user facilitation engine **460** as described in FIG. 4.

[0088] The sidebar **515** may also include one or more combinatory headings, such as starred combinatory heading **506**. A combinatory heading may aggregate one or more messages from one or more chat channels, according to a predetermined criterion. The combinatory headings may include a link that, in response to a user command, cause the client device to display one or more messages in the chat window **550**. The messages may be gathered from one or more chat channels, such as the chat channel **504a-b**, and displayed based on predetermined criteria. In FIG. 5, for example, the starred combinatory heading **506** may gather only those messages that have been marked by a user of the client device **340**. The marked messages may be stored at the client device **340**, and/or may be stored at the chat and video conference provider **310**. The link may cause the one or more processors included on the client device **340** to determine which messages are marked messages and cause them to be displayed in the chat window **550**. In some examples, the link may cause the client device **340** to send a signal to the chat and video conference provider **310**. The chat and video conference provider **310** may then determine which messages are marked messages and send information to the client device **340** to generate a display of the marked messages in the chat window **550**.

[0089] Other combinatory headings (and associated links and functionality) are also considered. Other examples may include an unread heading, an all files heading, a contact request heading, and others. As with the starred combinatory heading **506**, an associated link may cause the client device **340** and/or the chat and video conference provider **310** to determine which messages (if any) meet predetermined criteria associated with the combinatory heading and subsequently display those messages on the client device **340**.

[0090] The chat control dashboard **445** may display one or more control buttons and/or information regarding the chat channel currently being displayed on the client device. The control buttons may include links that mark a message (e.g., to mark it such that it is determined to be a marked message via the starred combinatory heading **406**), begin a video conference, schedule a meeting, create a video message, or other tasks. The chat control dashboard may also include a title of the

chat channel currently being displayed on the client device and/or a number of users with access to the chat channel. One of ordinary skill in the art would recognize many different possibilities and configurations.

[0091] The reply panel **555** may include an input field, where the user may cause the client device **340** to send a message to the chat channel. The input field may be accessed by a peripheral device such as a mouse, a keyboard, a stylus, or any other suitable input method. In some examples, the input field may be accessed by a touchscreen or other system built into the client device. In some examples, a notification may be sent from the client device **340** and/or the video conference provider **310** that indicates a response is being entered into the input field by the user. In other examples, no notification may be sent.

[0092] The reply dashboard **560** may include one or more buttons that, in response to a user command edit or modify a response input into the input field. For example, a record button may be provided, that allows the client device to capture audio and video. In other examples, there may be a share button that causes the client device to send the message to a different chat channel. In yet another example, there may be a reaction button that causes an image to be sent by the client device to the chat channel in response to a message posted in the chat channel.

[0093] In some examples, there may be one or more formatting buttons included on the reply dashboard **560**. The one or more formatting buttons may change the appearance of a replay entered in the input field. The user may thereby edit and customize their response in the input field before sending.

[0094] The reply dashboard **560** may include a send button **570**. The send button **570** may, in response to a user command, cause the client device **340** to send the contents of the input field (or “response”) to the chat channel. The client device **340** may then send the response to the chat and video conference provider **310**. The chat and video conference provider **310** may then send the response to the chat channel, which publishes the response for the users in the channel. The response may include image files such as JPEG, PNG, TIFF, or files in any other suitable format. The response may also include video files such as MPEG, GIF, or video files in any other suitable format. The response may also include text entered into the input field and/or other files attached to the message such as a PDF, DOC, or other file format.

[0095] The master chat panel **500** may include menus **535**. The menu **535** may include one or more buttons, which add functionality to the messages in the chat channels and/or the chat channels themselves. The menu **535** may be displayed in response to a user request on the client device. In the illustrated example, the user may have requested the menu **535** associated with a design team chat channel **504b**. The user may have requested the menu **535** from the client device using the cursor **540**. In some examples, mousing over a channel may cause the menu **535** to be displayed. In other examples, the menu **535** may be displayed in response to another user input (e.g., a right-click on the mouse while hovering over the design team chat channel **504**). In other examples, the menu **535** may be displayed by a user input occurring in another space rather than the design team chat channel **504**. For example, the menu **535** may be brought up by an input in a blank space of a chat window **550**. One of ordinary skill in the art would recognize many different possibilities.

[0096] The menu **535** may include options associated with a chat channel. Those options may include starting a video conference (or “meeting”), scheduling a meeting, adding or removing members from the chat channel, changing setting associated with notifications sent to the client device, no longer displaying the chat channel in sidebar **515**, recording a video message, and other such options. In some examples, the menu **535** may include options for a specific chat within the chat channel. For example, the menu **535** may include an option to mark a message as unread. As another example, the menu **535** may include an option to enable or disable facilitating user actions associated with chat sessions.

[0097] Although the menu **535** is illustrated as being opened from the design team chat channel **504b** on the sidebar **515**, the menu **535** may be opened from elsewhere. The menu **535** may display

the same functions or different functions depending on where it is opened. For example, there may be a button on the chat control dashboard **545** that causes the client device to open the menu **535**. Opening the menu **535** from the chat control dashboard **545** may only display functions related to a video meeting, for example. Functions related to a specific message in the chat may only be displayed in the menu **535** if the client device receives a user input on the specific message.

[0098] Now referring to FIG. 6, FIG. 6 shows an example master chat panel **600** displaying GUI elements and suggested reply message associated with an identified action item. The master chat panel **600** may be similar to the master chat panel **500** in FIG. 5 and therefore include some or all of the components and functionalities included therein. In this example, the chat window **650** includes a chat message **610**. The chat message **610** is from a user called “Alex”, not the local user logged in the master chat panel **600**. The chat message **610** asks the local user “when will you be free this afternoon?” The user facilitation engine **460** of the chat and video conference application can identify an action item that is to check the calendar based on the chat message **610**. The user facilitation engine **460** accesses the user's calendar and creates three GUI buttons for three available timeframes based on the user's calendar. GUI button **615** indicates 1-2 pm and can be activated (e.g., by clicking) to access the calendar and set up a meeting from 1 pm to 2 pm. Similarly, GUI button **620** indicates 3-3:30 pm and can be activated to set up a meeting from 3 pm to 3:30 pm in the calendar. GUI button **625** indicates 4-5 pm and can be activated to set up the meeting from 4 pm to 5 pm. The three GUI buttons **615**, **620**, and **625** are displayed in the reply panel **655** for the user to select based on the user's preference. The user does not need to go to the calendar to check the available times separately. When one GUI button is selected or the cursor is clicked on one GUI button, a reply message can be automatically composed based on the corresponding timeframe. For example, when the cursor **640** is clicked on the GUI button **620**, a reply message **630** is automatically composed to include 3-3:30 pm from the GUI button **620**. The user may edit the reply message **630** before sending it to the chat window **650**.

[0099] Now referring to FIG. 7, FIG. 7 shows an example master chat panel **700** displaying GUI elements with an identified action item in the chat window **750**. Chat message **705** is similar to chat message **610** in FIG. 6. The reply message **710** and the GUI button **715** may first appear in the reply panel **755** (not shown in FIG. 7), similar to the reply message **630** and the GUI buttons **615**-**625** in FIG. 6. For example, the user facilitation engine **460** identifies an action item to check the local user (e.g., Mike) calendar and generates a GUI button **715** indicating the meeting with Vijay scheduled at 3 pm on Sunday in the local user's calendar. The GUI button **715** can be activated to open the meeting schedule in the local user's calendar. The reply message **710** can be automatically composed based on the scheduled meeting. The local user can edit the reply message **710** and send it to the chat window **750**. Meanwhile, the GUI button **715** can also be displayed in the chat window **750** below the reply message **710**. The user may double check the schedule by clicking the GUI button **715**.

[0100] The chat message **720** asks the local user “Are you also planning to meet Ken?” The user facilitation engine **460** may identify an action item to check the local user's calendar for meeting with Ken. The user facilitation engine **460** may not find any meeting scheduled with Ken, but have suggested some available timeframes (not shown) based on the meeting time with Vijay, similar to the GUI buttons **615**-**625**. The user sends a reply message **725** “Let me arrange a meeting with Ken at 5 pm.” The user facilitation engine **460** can generate an action item to create a video meeting and generate a GUI button **730**. The GUI button **730** can be activated by clicking the cursor **640** at the GUI button. Then the user can create a meeting directly without navigating to a different application or GUI. The user facilitation engine **460** may automatically compose the content in different fields of the scheduling page (not shown) based on the reply messages **725**.

[0101] Now referring to FIG. 8, FIG. 8 shows an example master chat panel **800** displaying a chat summary message **855** in the chat window **850** of the design team chat channel. The master chat panel **800** may be similar to the master chat panel **500** in FIG. 5 and therefore include some or all

of the components and functionalities included therein. In this example, the chat window **850** of the design team channel includes a chat summary message **855**. The chat summary message **855** is generated by the user facilitation engine **460** as described in FIG. 4, for example from an external application (e.g., chat facilitation application **808**), and is only visible to the user associated with the client device **340**. The chat summary message **855** includes a list of next steps **805**, such as “Let me arrange a meeting with Ken at 5 pm” and “I’m gonna send the design material to JJulian,” which are chat messages from the chat session from 2:00 to 3:00 pm. The chat summary message **855** also includes a list of actionable items **810**, such as a GUI button **815** for creating a meeting with Ken and a GUI button **820** for sending material to JJulian. The GUI button **815** can be activated (e.g., by a mouse click) to invoke a user interface for scheduling a virtual meeting. The GUI button **820** can be activated (e.g., by a mouse click) to open a folder where the design material is saved.

[0102] Now referring to FIG. 9, FIG. 9 shows an example master chat panel **900** displaying a chat summary message **955** in an external application page **950**. The master chat panel **900** may be similar to the master chat panel **500** in FIG. 5 and therefore include some or all of the components and functionalities included therein. In this example, the external application page **950** includes a chat summary message **955**. The chat summary message **955** is generated by an external application (e.g., chat facilitation application **908**), and is only visible to the user associated with the client device **340**. The chat summary message **955** include information about the chat channel and time. The chat summary message **955** can be similar to the chat summary message **855** in FIG. 8, including a list of next steps and a list of actionable items. In this example, an actionable item is next to a corresponding next step. For example, next step **905** is from a chat message “Let me arrange a meeting with Ken at 5 pm.” A GUI button **915** can be activated (e.g., by a mouse click) to invoke a user interface for scheduling a virtual meeting. Also as an example, next step **910** is from a chat message “I’m gonna send the design material to Julian.” A GUI button **920** can be activated (e.g., by a mouse click) to open an email window for sending the design material to Julian. Julian’s email address can be automatically filled in the recipient field, and the subject line can be automatically filled. The design material can be attached automatically to the email.

[0103] Now referring to FIG. 10, FIG. 10 shows an example method **1000** for facilitating user actions during a chat session. The example method **1000** will be discussed with respect to the system **400** shown in FIG. 4; however, any suitable system for facilitating user actions may be used.

[0104] At block **1002**, a client device **340** receives a chat message via an online chat platform during a chat session. The online chat platform can be provided by the chat and video conference provider **310**. The client device **340** can access the online chat platform via a client-side application, for example a chat and video conference application **450** provided by the chat and video conference provider **310** and installed on the client device **340**. The chat message is from the chat session. The chat session can be a chat between different users in a chat channel. The chat message can be a text, an image, an audio clip, a video clip, or in any other suitable format. The chat message can be from a local user associated with the client device **340** via an input device of the client device **340**, for example, a mouse, a keyboard, a touchscreen, a microphone, a camera. Alternatively, the chat message is from another user in the chat session via another client device. The chat message is sent to chat and video conference application **450** installed on the client device **340** by the chat and video conference provider **310**.

[0105] At block **1004**, the client device **340** identifies an action item using an ML model at least based on the chat message. The ML model can be a classification model. The ML model can be initially trained by the chat and video conference provider **310** for classification. In some examples, the classification model is a rule-based model. In some examples, the ML model is trained in a supervised learning process, where category labels are provided so that the ML model learns certain features from the input data labeled with a particular action category. Alternatively, or

additionally, the ML model is trained during an unsupervised learning process, wherein the ML model learns certain features and identifies a corresponding action category. The chat and video conference provider **310** then transmits the trained ML model to the user facilitation engine **460** of the chat and video conference application **450** installed on the client device **340**. The user facilitation engine **460** implements the trained ML model to identify an action item from the chat message. In some examples, one or more chat messages prior to the chat message during the chat session are also used by the ML model to identify the action item.

[0106] In some examples, the chat and video conference provider **310** identifies an action item from the chat message. The chat and video conference provider **310** can include an action facilitation engine **430** configured to identify action items. The action facilitation engine **430** can implement a trained ML model from the model store **420**. For example, the ML model is a trained classification model, generally as described above.

[0107] At block **1006**, the client device **340** determines an application to execute the action item. In some examples, the application is a third-party application integrated with the chat and video conference application **450**. The chat and video conference application **450** can request access to other applications installed on the client device **340**. The chat and video conference application **450** can access these applications via APIs. In some examples, the application is the chat and video conference application **450** itself. The action item may be executed by a different functionality of the chat and video conference application **450**, such as video conferencing, phone calls, and any other suitable functions provided by the chat and video conference application **450**. The user facilitation engine **460** can determine what application installed on the client device **340** can execute the action indicated in the action item. For example, the action item is to check available times on calendar. The facilitation engine **360** determines which integrated application includes a calendar for the user. The integrated application can be an Outlook R calendar, Google® calendar, or any other calendar. In some examples, the chat and video conference provider **310** determines an application to execute the action item. In some examples, the ML model implemented by the action facilitation engine **430** includes a map between action categories and functionalities of applications or applications that can execute corresponding action items. In some examples, applications or functionalities of certain applications can be inferred from the action items identified by the ML model.

[0108] At block **1008**, the client device **340** generates one or more GUI elements associated with a functionality of the application corresponding to the action item. The user facilitation engine **460** can be configured to generate GUI elements based on the identified action items and associated applications or functionalities of certain applications. The one or more GUI elements can include an indication of the functionality of the application corresponding to the action item. The one or more GUI elements can be a button, a check box, a drop list, or in any other suitable formats. The one or more GUI element can include a description of the action item and an icon of the application that can execute the action indicated in the action item. The one or more GUI elements can trigger a functionality of the application that can execute the action indicated by the action item. In some examples, the action facilitation engine **430** identifies the action items and determines the associated applications or functionalities of the applications that can execute the action items. The action facilitation engine **430** can transmit the information about the identified action items and associated application (or functionality thereof) to the user facilitation engine **460**. The user facilitation engine **460** then generates one or more GUI elements associated with a functionality of the application corresponding to the action item.

[0109] At block **1010**, the client device **340** causes the one or more GUI elements to be displayed in a GUI of the chat session. In some examples, the chat message is from another user, the one or more GUI elements are generated for the local user associated with client device **340** to facilitate replying to the chat message received. The one or more GUI elements can then be displayed in a reply panel for the user's reference, for example GUI buttons **615-625** in FIG. **6**. In some examples,

the chat message is from the local user associated with the client device **340**, the one or more GUI elements are generated to facilitate an action identified in the local user's chat message. The one or more GUI elements can then be displayed under the chat message in the chat window, for example GUI button **730** in FIG. 7.

[0110] At block **1012**, the client device **340** receives a triggering signal for invoking the functionality of the application from a GUI element of the one or more GUI elements. The GUI element can be activated by a mouse click, a finger press, a stylus, a keystroke, or any suitable means. A triggering signal can be generated by the GUI element being activated. The triggering signal is to invoke the functionality of the application associated with the GUI element being activated. In some examples, the user facilitation engine **460** of the client device **340** can automatically compose a chat message based on the functionality of the application invoked by the triggering signal, for example the reply message **630** in FIG. 6.

[0111] At block **1014**, the client device **340** launches the application to invoke the functionality in response to receiving the triggering signal. The application can be automatically launched to invoke the function by activating a corresponding GUI element. A user interface or a window associated with the functionality can be displayed on the client device **340**. For example, a scheduling page can pop up for the user to schedule a meeting identified in the chat message.

[0112] In some examples, the user facilitation engine **460** identifies several action items from multiple chat messages during the chat session and creates corresponding GUI elements for the action items. Some of the GUI elements are activated to execute the actions indicated in corresponding action items during the chat session. Some of the GUI elements are not activated during the chat session. The user facilitation engine **460** can also identify some action items without corresponding GUI elements, but they include actions to be taken by the user. In some examples, the user facilitation engine **460** creates a summary for the chat session after the chat session is ended. The summary can be sent to the chat window of the chat session. Alternatively, the summary can be sent to a page associated with the integrated facilitation application. Still alternatively, the summary can be sent in an email or other communication mean. The summary includes quotes of chat messages indicating user actions. Some of these quoted chat messages correspond to a GUI element, either activated or not activated yet. Some of these quoted chat messages do not have corresponding GUI elements, indicating that the actions indicated in the quoted chat message may not be carried out by an application in the client device **340**. The summary may also include a feedback request. The user facilitation engine **460** can create a blank field at the end of the summary message for soliciting feedback from the user. Alternatively, or additionally, the facilitation engine can include radio buttons for action item identified in the summary for user feedback. It can be understood that other formats of feedback inputs are also possible.

[0113] Now referring to FIG. 11, FIG. 11 shows an example computing device **1100** suitable for use in example systems or methods for facilitating user actions for virtual interactions. The example computing device **1100** includes a processor **1110** which is in communication with the memory **1120** and other components of the computing device **1100** using one or more communications buses **1102**. The processor **1110** is configured to execute processor-executable instructions stored in the memory **1120** to perform one or more methods for facilitating user actions for virtual interactions, such as part or all of the example method **1000**, described above with respect to FIG. 10. The computing device, in this example, also includes one or more user input devices **1150**, such as a keyboard, mouse, touchscreen, video input device (e.g., one or more cameras), microphone, etc., to accept user input. The computing device **1100** also includes a display **1140** to provide visual output to a user. The computing device **1100** may also include a chat and video conference software **1160**. The chat and video conference software **1160** may include a chat client, a video conference platform, and any other software to enable communication from a first user to a second user.

[0114] The computing device **1100** also includes a communications interface **1130**. In some examples, the communications interface **1130** may enable communications using one or more networks, including a local area network (“LAN”); wide area network (“WAN”), such as the Internet; metropolitan area network (“MAN”); point-to-point or peer-to-peer connection; etc. Communication with other devices may be accomplished using any suitable networking protocol. For example, one suitable networking protocol may include the Internet Protocol (“IP”), Transmission Control Protocol (“TCP”), User Datagram Protocol (“UDP”), or combinations thereof, such as TCP/IP or UDP/IP.

[0115] While some examples of methods and systems herein are described in terms of software executing on various machines, the methods and systems may also be implemented as specifically-configured hardware, such as field-programmable gate array (FPGA) specifically to execute the various methods according to this disclosure. For example, examples can be implemented in digital electronic circuitry, or in computer hardware, firmware, software, or in a combination thereof. In one example, a device may include a processor or processors. The processor comprises a computer-readable medium, such as a random access memory (RAM) coupled to the processor. The processor executes computer-executable program instructions stored in memory, such as executing one or more computer programs. Such processors may comprise a microprocessor, a digital signal processor (DSP), an application-specific integrated circuit (ASIC), field programmable gate arrays (FPGAs), and state machines. Such processors may further comprise programmable electronic devices such as PLCs, programmable interrupt controllers (PICs), programmable logic devices (PLDs), programmable read-only memories (PROMs), electronically programmable read-only memories (EPROMs or EEPROMs), or other similar devices.

[0116] Such processors may comprise, or may be in communication with, media, for example one or more non-transitory computer-readable media, that may store processor-executable instructions that, when executed by the processor, can cause the processor to perform methods according to this disclosure as carried out, or assisted, by a processor. Examples of non-transitory computer-readable medium may include an electronic, optical, magnetic, or other storage device capable of providing a processor, such as the processor in a web server, with processor-executable instructions. Other examples of non-transitory computer-readable media include a floppy disk, CD-ROM, magnetic disk, memory chip, ROM, RAM, ASIC, configured processor, all optical media, all magnetic tape or other magnetic media, or any other medium from which a computer processor can read. The processor, and the processing, described may be in one or more structures, and may be dispersed through one or more structures. The processor may comprise code to carry out methods (or parts of methods) according to this disclosure.

[0117] The foregoing description of some examples has been presented only for the purpose of illustration and description and is not intended to be exhaustive or to limit the disclosure to the precise forms disclosed. Numerous modifications and adaptations thereof will be apparent to those skilled in the art without departing from the spirit and scope of the disclosure.

[0118] Reference herein to an example or implementation means that a particular feature, structure, operation, or other characteristic described in connection with the example may be included in at least one implementation of the disclosure. The disclosure is not restricted to the particular examples or implementations described as such. The appearance of the phrases “in one example,” “in an example,” “in one implementation,” or “in an implementation,” or variations of the same in various places in the specification does not necessarily refer to the same example or implementation. Any particular feature, structure, operation, or other characteristic described in this specification in relation to one example or implementation may be combined with other features, structures, operations, or other characteristics described in respect of any other example or implementation.

[0119] Use herein of the word “or” is intended to cover inclusive and exclusive OR conditions. In other words, A or B or C includes any or all of the following alternative combinations as

appropriate for a particular usage: A alone; B alone; C alone; A and B only; A and C only; B and C only; and A and B and C.

Claims

1. A method comprising: accessing, by a client device during a chat session, a chat message; identifying, by the client device during the chat session, an action item for a chat participant associated with the client device based at least on the chat message using a trained machine learning (ML) model; generating, by the client device during the chat session, one or more graphical user interface (GUI) elements associated with a functionality of an application to execute the action item; causing, by the client device during the chat session, the one or more GUI elements to be displayed in a GUI of the client device; executing, by the client device during the chat session, the application to invoke the functionality in response to receiving a triggering signal; and generating, by the client device after the chat session, a summary of action items, comprising a plurality of action items.
2. The method of claim 1, further comprising receiving, by the client device during the chat session, the triggering signal for invoking the functionality of the application corresponding to a GUI element of the one or more GUI elements, wherein the triggering signal is generated by the GUI element being activated.
3. The method of claim 1, further comprising: receiving user feedback data related to the one or more GUI elements associated with the functionality of the application corresponding to the action item; and fine-tuning the trained ML model based on the user feedback data.
4. The method of claim 1, wherein the application comprises a calendar application, wherein the method further comprises: accessing calendar data in the calendar application associated with the chat participant; and generating the one or more GUI elements associated with the calendar application to execute the action item.
5. The method of claim 1, further comprising: accessing user data associated with the application; and generating a draft response message in reply to the chat message based on the user data using a generative artificial intelligence (AI) model.
6. The method of claim 1, wherein the plurality of action items comprises a first list of action items the user has taken action on during the chat session and a second list of action items the user has yet to take action on after the chat session.
7. The method of claim 1, wherein the summary of action items further comprises a list of chat messages indicating the plurality of action items and a plurality of GUI elements that the user has yet to take action on.
8. A system comprising: a communications interface; a non-transitory computer-readable medium; and one or more processors communicatively coupled to the communications interface and the non-transitory computer-readable medium, the one or more processors configured to execute processor-executable instructions stored in the non-transitory computer-readable medium to: access a chat message during a chat session; identify an action item for a chat participant based at least on the chat message using a trained machine learning (ML) model; generate one or more graphical user interface (GUI) elements associated with a functionality of an application to execute the action item; cause the one or more GUI elements to be displayed in a GUI of a client device; execute the application to invoke the functionality in response to receiving a triggering signal; and generate a summary of action items, comprising a plurality of action items.
9. The system of claim 8, wherein the one or more processors are configured to execute further processor-executable instructions stored in the non-transitory computer-readable medium to: receive a triggering signal for invoking the functionality of the application corresponding to a GUI element of the one or more GUI elements, wherein the triggering signal is generated by the GUI element being activated.

10. The system of claim 8, wherein the one or more processors are configured to execute further processor-executable instructions stored in the non-transitory computer-readable medium to: receive user feedback data related to the one or more GUI elements associated with the functionality of the application corresponding to the action item; and fine-tune the trained ML model based on the user feedback data.

11. The system of claim 8, wherein the application comprises a calendar application, wherein the one or more processors are configured to execute further processor-executable instructions stored in the non-transitory computer-readable medium to: access calendar data in the calendar application associated with the chat participant; and generate the one or more GUI elements associated with the calendar application to execute the action item.

12. The system of claim 8, wherein the one or more processors are configured to execute further processor-executable instructions stored in the non-transitory computer-readable medium to: access user data associated with the application; and generate a draft response message in reply to the chat message based on the user data using a generative artificial intelligence (AI) model.

13. The system of claim 8, wherein the plurality of action items comprises a first list of action items the user has taken action on during the chat session and a second list of action items the user has yet to take action on after the chat session.

14. The system of claim 8, wherein the summary of action items further comprises a list of chat messages indicating the plurality of action items and a plurality of GUI elements that the user has yet to take action on.

15. A non-transitory computer-readable medium comprising processor-executable instructions configured to cause one or more processors to: access a chat message during a chat session; identify an action item for a chat participant based at least on the chat message using a trained machine learning (ML) model; generate one or more graphical user interface (GUI) elements associated with a functionality of an application to execute the action item; cause the one or more GUI elements to be displayed in a GUI of a client device; execute the application to invoke the functionality in response to receiving a triggering signal; and generate a summary of action items, comprising a plurality of action items.

16. The non-transitory computer-readable medium of claim 15, further comprising processor-executable instructions configured to cause one or more processors to: receive a triggering signal for invoking the functionality of the application corresponding to a GUI element of the one or more GUI elements, wherein the triggering signal is generated by the GUI element being activated.

17. The non-transitory computer-readable medium of claim 15, further comprising processor-executable instructions configured to cause one or more processors to: receive user feedback data related to the one or more GUI elements associated with the functionality of the application corresponding to the action item; and update the trained ML model based on the user feedback data.

18. The non-transitory computer-readable medium of claim 15, wherein the application comprises a calendar application, further comprising processor-executable instructions configured to cause one or more processors to: access calendar data in the calendar application associated with the chat participant; and generate the one or more GUI elements associated with the calendar application to execute the action item.

19. The non-transitory computer-readable medium of claim 15, further comprising processor-executable instructions configured to cause one or more processors to: access user data associated with the application; and generate a draft response message in reply to the chat message based on the user data using a generative artificial intelligence (AI) model.

20. The non-transitory computer-readable medium of claim 15, wherein the summary of action items further comprises a list of chat messages indicating the plurality of action items, and wherein the plurality of action items comprises a first list of action items the user has taken action on during

the chat session and a second list of action items the user has yet to take action on after the chat session.
